



Agile Compass: A Tool for Identifying Maturity in Agile Software-Development Teams

Rafaela Mantovani Fontana, Sheila Reinehr, and Andreia Malucelli

RESEARCHERS HAVE PROPOSED many agile software-development maturity models to provide guidelines for work process improvements.¹ Most of them describe various maturity levels and the processes that teams should adopt to achieve each one.² A team then uses the model to assess its current level and identify the practices that would guide them to maturity.

Agile software-development methods present a challenge to using maturity models because their practices have been highly customized for specific contexts.³ If agile teams don't usually rely on standard processes, how could they use a maturity model to identify where they are and how to move forward?

To answer this question, we researched how agile software-development teams naturally evolve (see the sidebar "Our Research Approach"). We analyzed nine teams' evolution of practices and found that the process was, as we expected, idiosyncratic. Each team adopted practices based on its circumstances and improved the practices based on the challenges it faced.

Using this research, we designed the Agile Compass, a checklist (see the Web extra at https://s3.amazonaws.com/ieeecs.cdn.csdl.public/mags/so/2015/06/extras/mso201506_AnAgile_s1.pdf) that lets agile-development teams identify where they are on the road to agile

maturity without prescribing practices or adoption levels.

The Road to Maturity

We found that teams accomplished agile maturity via a dynamic evolution based on the pursuit of specific outcomes. We identified the various types of outcomes and organized them in a framework (see Figure 1).

Identifying the outcomes a team has and hasn't accomplished doesn't provide an exact roadmap for the future, as prescribed processes or maturity levels do. Instead, this process gives a general direction as to which further improvements should occur, leaving room for context-specific endeavors.

We identified seven outcome categories:

- *Practices learning* includes the learning-related outcomes teams pursue when they decide to change how they work.
- *Team conduct* describes how the team—the central role in agile maturity—evolves.
- *Pace of deliveries* describes the delivery-related outcomes the team progressively accomplishes while evolving.
- *Features disclosure* shows how the dynamics of the definition of software requirements change over time.





OUR RESEARCH APPROACH

We interviewed nine agile software-development teams (see Table A). We analyzed each transcribed interview¹ to identify how agile practices evolved and are expected to evolve for each team. We then identified evidence that led to the outcomes described in our progressive-outcomes framework (see Figure 1 in the main article). For a more detailed description of the research process, see the “Agile Compass Research Approach” Web extra at https://s3.amazonaws.com/ieeecs.cdn.csdl.public/mags/so/2015/06/extras/mso201506_AnAgile_s2.png.

We developed our Agile Compass checklist based on our outcomes framework. For example, we interviewed teams and received these responses that indicated they used these approaches to develop high-level source code:

- Company A performed pair programming, refactored code, and cared about the code.
- Company B, team 2, integrated code daily.
- Company D performed pair programming, cleaned code, and refactored code.
- Company E, team 1, conducted code reviews, provided support from an architecture team, performed pair programming, defined a best-practices checklist, gave people time to study coding techniques, and encouraged them to study coding techniques.
- Company E, team 2, didn't automate deployment,

refactored code with regular builds, and performed pair programming.

- Company F performed pair programming, used test-driven development, and conducted informal code reviews.

Analyzing this information and taking into account the context of each team's project and our observation of each team, we created the part of the Agile Compass (see the Web extra at https://s3.amazonaws.com/ieeecs.cdn.csdl.public/mags/so/2015/06/extras/mso201506_AnAgile_s1.pdf) pertaining to the development of high-level source code. We described this outcome by saying, “Coding is an important activity, and there are initiatives to guarantee the code is clear and robust, using best practices available.” We then summed up those practices in the form of two statements to which companies could agree, if applicable: “We care about our code” and “We endeavor initiatives to guarantee our code is fine, such as reviews, refactoring, pair programming, or others.”

All information we gathered that supported each outcome is in our “Map of Evidence” Web extra at https://s3.amazonaws.com/ieeecs.cdn.csdl.public/mags/so/2015/06/extras/mso201506_AnAgile_s3.png.

Reference

1. R.M. Fontana et al., “Progressive Outcomes: A Framework for Maturing in Agile Software Development,” *J. Systems and Software*, vol. 102, 2015, pp. 88-108.

TABLE A

Profiles of interviewed development teams

	Team size	Company size (no. of employees)	Years since adopting agile development	Purpose for developing software
Company A	8	150	1.3	Software for the company's own use and for external customers
Company B, team 1	5	200	5	Software built on demand for external customers
Company B, team 2	7	200	6	Software packages and embedded systems
Company C	20	15,000	3	Customization or adaptation of existing software
Company D (distributed)	5	17	6	Software for the company's own use or built on demand
Company E, team 1	12	90	5	Customization or adaptation of existing software
Company E, team 2 (second site)	6	150	5	Software for the company's own use or built on demand
Company F	8	150	4	Software for the company's own use or built on demand
Company G (government organization)	11	10,000	0.4	Customization or adaptation of existing software

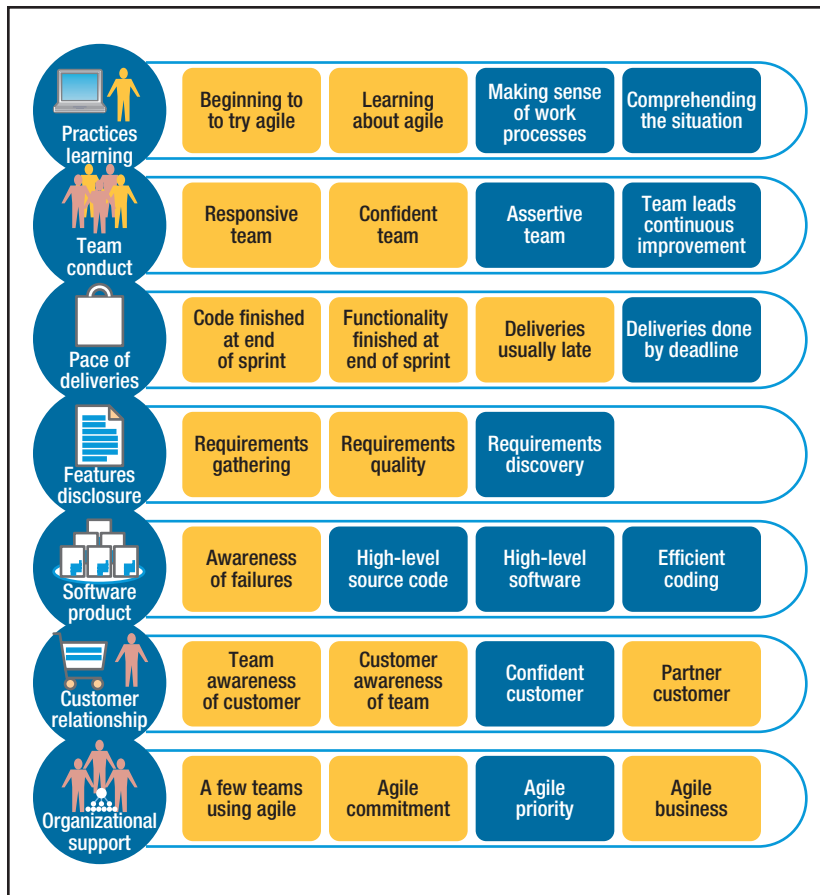


FIGURE 1. The progressive-outcomes framework. The circles represent outcome categories; the boxes are various development outcomes. The blue boxes are outcomes that indicate team maturity.

- *Software product* shows how the quality of the team's software evolves.
- *Customer relationship* describes the team's changing relationship with the customer.
- *Organizational support* represents how organizational support for agile development changes during the maturity process.

We lack evidence that teams either pursue all outcomes or follow a predefined sequence of outcomes in the maturing process. However, we have clear evidence of the outcomes that mature teams have achieved, which appear as the blue boxes in Figure 1. The other boxes are outcomes that teams naturally pursue during their agile evolution but that don't always characterize maturity.

Where Are We?

In the cases we studied, we identified dozens of practices that contribute to the accomplishment of outcomes (see Table 1) and used this information to develop the Agile Compass.

The Agile Compass briefly describes each outcome in an agile-development process and offers several ways to identify whether it was accomplished. Development teams can use the tool to discuss and determine which outcomes they've accomplished and which still need work, and where they must invest additional effort to improve the process.

Using the Agile Compass doesn't necessarily give a final, definitive picture of a team's progress toward maturity because the situation can change

TABLE 1

Agile Compass research profile.*

Outcome category	No. of pieces of information from interviews showing development teams pursuing each outcome
Practices learning	65
Team conduct	79
Pace of deliveries	38
Features disclosure	48
Software product	83
Customer relationship	35
Organizational support	25

* The study involved nine teams (with 25 practitioners) and 27 identified outcomes.

over time. For example, when a variable—such as a team member or a technology used to build the software—changes, the team might have to start pursuing its outcomes again. In addition, once developers in a team accomplish mature outcomes, their challenge is to continuously improve work processes to sustain these outcomes in a changing business environment.

As the business environment changes, management must observe its development teams, use some simple metrics to evaluate them, and make the necessary adjustments as quickly as possible to enable continuous improvement.⁴ The improvement is based not on prescribed processes but on the pursuit of outcomes. This approach places the agile team in a central role and supports the con-

stant learning of emerging practices for specific development contexts. 

References

1. M. Leppänen, “A Comparative Analysis of Agile Maturity Models,” *Information Systems Development: Reflections, Challenges and New Directions*, R. Pooley et al., eds., Springer, 2013, pp. 329–343.
2. R.M. Fontana et al., “Progressive Outcomes: A Framework for Maturing in Agile Software Development,” *J. Systems and Software*, vol. 102, 2015, pp. 88–108.
3. D. Bustard, G. Wilkie, and D. Greer, “The Maturation of Agile Software Development Principles and Practice: Observations on Successive Industrial Studies in 2010 and 2012,” *Proc. 20th Ann. IEEE Int’l Conf. and Workshops Eng. Computer-Based Systems (ECBS 13)*, 2013, pp. 139–146.
4. M. Walter et al., “From Sprints to Lean Flow: Management Strategies for Agile Improvement,” *Agile Processes in Software Engineering and Extreme Programming*, C. Lassenius, T. Dingsøyr, and M. Paasi-vaara, eds., Springer, 2015, pp. 310–318.

RAFAELA MANTOVANI FONTANA is a doctoral student in software engineering at the Pontifical Catholic University of Paraná and a professor of software engineering at the Federal University of Paraná. Contact her at rafaela.fontana@ufpr.br.

SHEILA REINEHR is a doctoral supervisor and professor of software engineering at the Pontifical Catholic University of Paraná. Contact her at sheila.reinehr@pucpr.br.

ANDREIA MALUCELLI heads the postgraduate program in informatics at the Pontifical Catholic University of Paraná. Contact her at malu@ppgia.pucpr.br.



See www.computer.org/software-multimedia for multimedia content related to this article.



handles the details
so you don't have to!

- Professional management and production of your publication
- Inclusion into the IEEE Xplore and CSDL Digital Libraries
- Access to CPS Online: Our Online Collaborative Publishing System
- Choose the product media type that works for your conference:
Books, CDs/DVDs, USB Flash Drives, SD Cards, and Web-only delivery!

Contact CPS for a Quote Today!

www.computer.org/cps or cps@computer.org



IEEE  computer society